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Medial-lateral position of hand changes shoulder muscle synergy during the push-up exercises

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***Abstract*—The aim of this study was to compare muscle synergy among different medial-lateral positions of the hand during the push-up exercises: narrow, normal, and wide. Fifteen fitness athletes participated (weight: 68.35±7.18 kg, height: 175±3.40 cm, age: 24.50±7.5 years, and experience: 12±3 years). The kinematics data were recorded via ten motion captures. The EMG activities were collected for selective muscles: the anterior deltoid, middle deltoid, posterior deltoid, infraspinatus, upper pectoralis major, middle pectoralis major, lower pectoralis major, latissimus dorsi, triceps lateral, triceps medial, biceps brachii, upper trapezius, middle trapezius, and lower trapezius muscles. An NMF (Non-negative matrix factorization) algorithm was applied to extract muscle synergy from EMG data for three different medial-lateral hand positions of the push-up exercises. The scalar product (SP) is a similarity index that was used to quantify the similarity of muscle synergies among different conditions. Based on the results, two synergies were extracted from the EMG of 14 muscles. Synergy 1 activated at the lowest position. While synergy 2 contains the before and after the lowest position. Therefore, the most active muscles correspond to shoulder and elbow extension/flexion. Each synergy was controlled groups of muscles with similar biomechanical action and associated with the push-up phases. The minimum similarity of synergy coefficient was observed between normal and wide positions for both synergy 1 (SP=0.88) and synergy 2 (SP=0.79). The minimum similarity of synergy weight was observed between normal and wide positions for both synergy 1 (SP=0.91) and synergy 2 (SP=0.84). We have demonstrated the presence of muscle coordination in the push-up and examined their relationship with biomechanical variables. These results indicate how athletes and coaches can improve their performances by understanding the muscles synergies.**

Keywords-component; Push-up; Muscle; Synergy; Shoulder; Coordination

I. INTRODUCTION

Push-up is an exercise that various groups such as physiotherapists, coaches, and trainers use to boost the upper body [1]. There are different types of push-ups that involve different muscles [2]. Changes in the positions of the hands can lead to changes in muscle activations [3]. On the other hand, coordination among muscles was controlled by the central nervous system (CNS) [4]–[7]. The CNS controls muscle coordination based on input information that came from many factors such as kinematics and kinetics variables [8]. Therefore, it is important to reveal the relationship between kinematics variables and muscle coordination [9].

Muscle synergy is a technique to quantify muscle coordination by decreasing the dimensionality of muscle control [10]. Muscle synergy provides information about how the CNS controls the motions [11]. Usually, muscle synergy was used as an indicator to evaluate the performance of subjects [12]. It was used to analyze sports activities [13]–[17], patients after stroke [18], and robots [19].

In recent years various studies evaluated the effect of push-up variances on muscle activation [20]. Many researchers compared different conditions of push-up and positions of hands [21]. However, there is still a lack of information about muscle coordination during the push-up exercise and how the CNS organizes the muscle coordination in various positions of hands. Therefore, the goals of this study were to (1) extract muscle synergy from the EMG signals of selective muscles and (2) compare muscle synergy among three different medial-lateral positions of hands: narrow, normal, and wide.

II. METHODS AND MATERIALS

A. Subjects

Fifteen healthy fitness athletes participated in this study. All subjects read and signed the letter of satisfaction form before experiments [22], [23]. All subjects were free from any musculoskeletal injuries. Table 1 shows the characteristics of subjects includes weight, age, and height [24], [25]. Table 1 presents the anthropometrics and experience information for the subjects.

Table 1. Anthropometrics and experience information for the subjects.

	Mean (± SD)
Age (years)	24.50 (7.50)
Weight (kg)	68.35 (7.18)
Height (cm)	175 (3.40)

All values are means ± SD

B. Instrumentation

Ten high-speed motion capture (Vicon MX, Oxford, UK, 100 Hz) were used to record the 3D positions of surface

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reflective markers during the push-up exercises [26][23], [27]. The fourteen channels of the Myon EMG system (made in Switzerland) with a sampling frequency of 1000 Hz were used to record muscle activation signals of the anterior deltoid, middle deltoid, posterior deltoid, infraspinatus, upper pectoralis major, middle pectoralis major, lower pectoralis major, latissimus dorsi, triceps lateral, triceps medial, biceps brachii, upper trapezius, middle trapezius, and lower trapezius muscles. The instructions of EMG electrodes placements were according to the recommendation of Cram and Kasman [28]. Before attaching the electrodes, the skin shaving and cleaning with alcohol were accomplished to decrease the signal noises [29]. Figure 1 shows the electrodes and markers placements in the motion analysis lab.

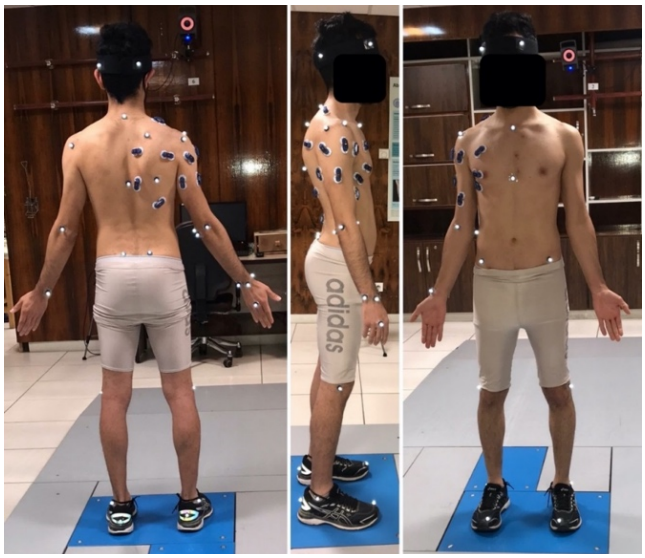


Fig. 1. The markers and EMG electrodes placements. The EMG electrodes attached on the anterior deltoid, middle deltoid, posterior deltoid, infraspinatus, upper pectoralis major, middle pectoralis major, lower pectoralis major, latissimus dorsi, triceps lateral, triceps medial, biceps brachii, upper trapezius, middle trapezius, and lower trapezius muscles.

C. Experimental Setup

After fifteen minutes of warming up and stretching exercises, the subjects were informed about the experimental procedure [30], [31]. Each subject executed three different types of push-ups. The push-up types were differed based on medial-lateral positions of hands [3]. Tables 2 presents the characteristics of each push-up type. Each type of push-up was performed five times by each subject.

Table 2: Categorization of the different medial-lateral hand positions.

	Medial-Lateral distance between hands	Superior-Inferior position of the hands	Rotation position of the hands
Narrow	0 cm	In front of the face	0° degree
Normal	The distance between left and right shoulder markers	In front of the face	0° degree
Wide	2*(the distance between left and right shoulder markers)	In front of the face	0° degree

D. Data Analysis

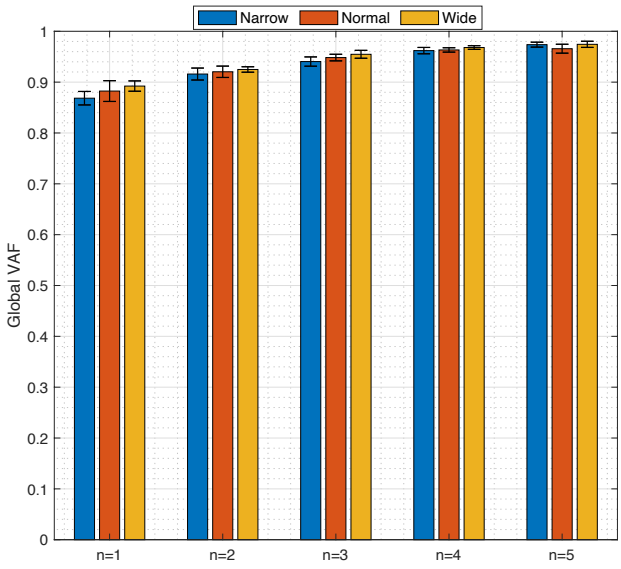
The EMG data were rectified and filtered by a 4th order, Butterworth, low pass filter with a cut-off frequency of 6 Hz. The EMG and motion capture data were normalized to facilitate analysis and comparisons [32]. The non-negative matrix factorization (NMF) method was used to extract muscle synergies from the EMG data. The global variance accounted for (Global_VAF) was used to calculate the number of muscle synergies. We applied the least number of synergies that accounted for >90% of the global_VAF according to our previous study [15][33]. The local_VAF for each muscle was calculated to ensure that each muscle activity pattern was well reconstructed by the NMF algorithm. The adoption level was defined at 0.75 for local_VAF for each muscle according to our previous study [15]. The scalar product (SP) which is the similarity index between two synergies was calculated to compare muscle synergy among various conditions according to the study of Cheung [34].

E. Statistics

The mean and standard deviations were used to state the results. The one-way ANOVA was applied to compare the three different conditions of push-ups. The significant level was defined at 0.05. All statistical analysis was performed in MATLAB 2020a software.

III. Results

Fig. 2 presents the mean and standard deviation of global_VAF for the different number of synergies extracted from the EMG signal for three types of push-ups: narrow, normal, and wide. These results indicated two muscle synergies are required to reconstruct the EMG data by calculated muscle synergies for all types of push-ups: narrow, normal, and wide.



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Fig. 2. The mean and standard deviation of global_VAF for the different number of synergies extracted from the EMG signal for three types of push-ups: narrow, normal, and wide.

Fig. 3 represents the mean and standard deviation of local_VAF for the anterior deltoid, middle deltoid, posterior deltoid, infraspinatus, upper pectoralis major, middle pectoralis major, lower pectoralis major, latissimus dorsi, triceps lateral, triceps medial, biceps brachii, upper trapezius, middle trapezius, and lower trapezius muscles for three types of push-ups: narrow, normal, and wide. Based on these results, the local_VAF for all muscles and push-up types were greater than 0.75 that shows the reliability of the muscle synergies.

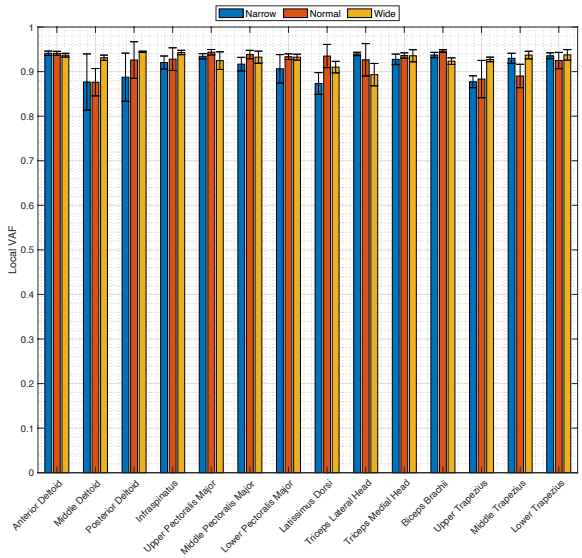


Fig. 3. The mean and standard deviation of local_VAF for the anterior deltoid, middle deltoid, posterior deltoid, infraspinatus, upper pectoralis major, middle pectoralis major, lower pectoralis major, latissimus dorsi, triceps lateral, triceps medial, biceps brachii, upper trapezius, middle trapezius, and lower trapezius muscles for three types of push-ups: narrow, normal, and wide.

Fig. 4 presents the mean and standard deviation of synergy weights and synergy coefficients for the anterior deltoid, middle deltoid, posterior deltoid, infraspinatus, upper pectoralis major, middle pectoralis major, lower pectoralis major, latissimus dorsi, triceps lateral, triceps medial, biceps brachii, upper trapezius, middle trapezius, and lower trapezius muscles for three types of push-ups: narrow, normal, and wide. Synergy 1 activated at the lowest position. While synergy 2 contains the before and after the lowest position.

Table 3 presents the similarity index between narrow, normal, and wide for synergy coefficient, synergy weight, and fourteen muscles: the anterior deltoid, middle deltoid, posterior deltoid, infraspinatus, upper pectoralis major, middle pectoralis major, lower pectoralis major, latissimus dorsi, triceps lateral, triceps medial, biceps brachii, upper trapezius, middle trapezius, and lower trapezius muscles. The minimum similarity of synergy coefficient was observed between normal and wide positions for both synergy 1 (SP=0.88) and synergy 2 (SP=0.79). The minimum similarity of synergy weight was observed between normal and wide positions for both synergy 1 (SP=0.91) and synergy 2 (SP=0.84).

Table 3: Similarity index between narrow, normal, and wide for synergy coefficient, synergy weight, and fourteen muscles: the anterior deltoid, middle deltoid, posterior deltoid, infraspinatus, upper pectoralis major, middle

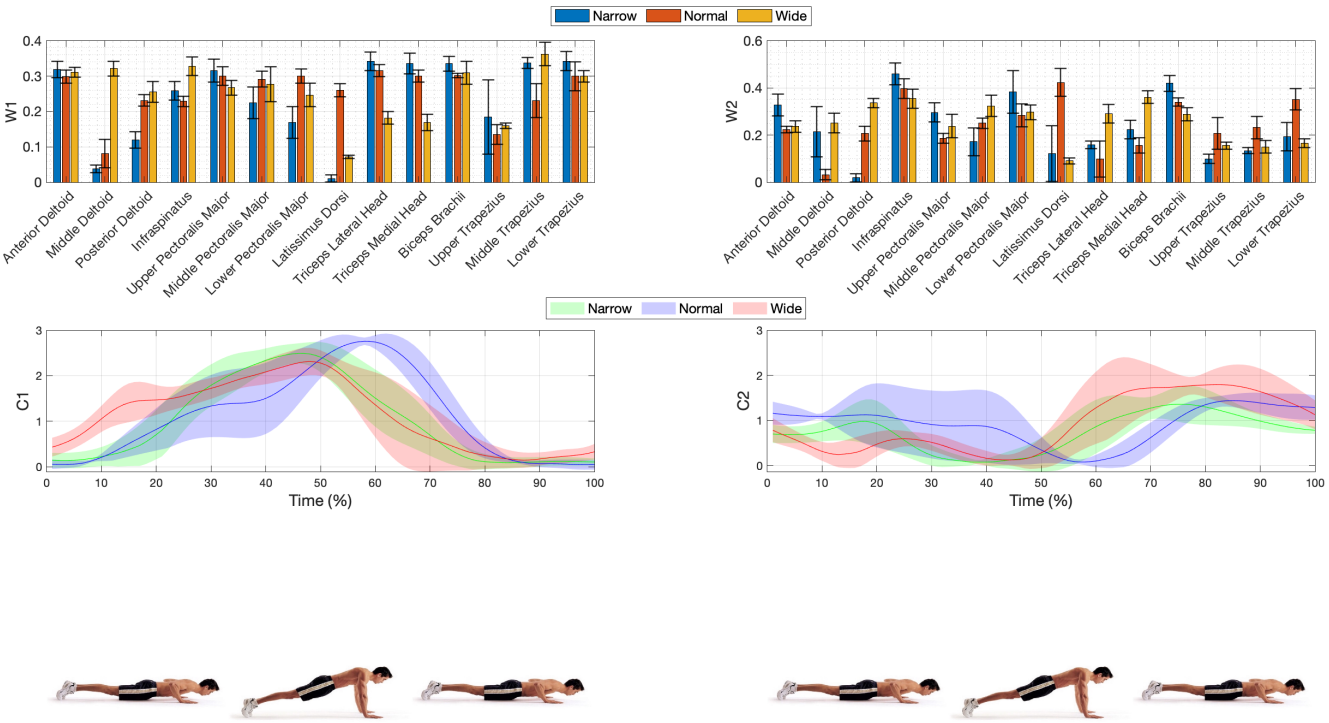


Fig. 4. The mean and standard deviation of synergy weights and synergy coefficients for the anterior deltoid, middle deltoid, posterior deltoid, infraspinatus, upper pectoralis major, middle pectoralis major, lower pectoralis major, latissimus dorsi, triceps lateral, triceps medial, biceps brachii, upper trapezius, middle trapezius, and lower trapezius muscles for three types of push-ups: narrow, normal, and wide.

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pectoralis major, lower pectoralis major, latissimus dorsi, triceps lateral, triceps medial, biceps brachii, upper trapezius, middle trapezius, and lower trapezius muscles.

		Narrow VS Normal	Normal VS Wide	Narrow VS Wide
Synergy Coefficient	Synergy 1	0.91	0.88	0.96
	Synergy 2	0.85	0.79	0.95
Synergy Weight	Synergy 1	0.94	0.91	0.92
	Synergy 2	0.86	0.84	0.89
Muscles	DELT1	0.99	1.00	0.99
	DELT2	0.53	0.96	0.74
	DELT3	0.84	0.98	0.72
	INFSP	1.00	0.97	0.97
	PECM1	0.98	0.99	1.00
	PECM2	1.00	0.99	0.98
	PECM3	0.92	0.99	0.96
	LAT2	0.89	0.99	0.84
	TRILat	0.99	0.76	0.84
	TRImed	0.99	0.80	0.86
	BIClong	1.00	1.00	0.99
	TRP1	0.88	0.98	0.96
	TRP3	0.92	0.92	1.00
	TRP4	0.94	0.94	1.00

IV. Discussion

The aims of the current study were to (1) extract muscle synergy from the EMG signals of selective muscles and (2) compare muscle synergy among three different medial-lateral positions of hands: narrow, normal, and wide. The NMF algorithm was used to compare muscle synergy among various positions of the hands during standard push-up exercise. Based on the results, two synergies were extracted from the EMG of 14 muscles. Synergy 1 activated at the lowest position. While synergy 2 contains the before and after the lowest position. Therefore, the most active muscles correspond to shoulder and elbow extension/flexion. Each synergy was controlled groups of muscles with similar biomechanical action and associated with the push-up phases. We have demonstrated the presence of muscle coordination in the push-up and examined their relationship with biomechanical variables. These results indicate how athletes and coaches can improve their performances by understanding the muscles synergies. The minimum similarity of both synergy coefficient and synergy weight were observed between normal and wide positions for both. These results indicated that maximum differences for CNS strategy to control the coordination of muscles for the normal and wide conditions.

V. Conclusion

This study is the first attempt that provides valuable information about the muscle synergy differences for various types of push-ups. This study was revealed how the CNS organized and controlled the motor coordination when medial-lateral positions (narrow, normal, and wide) of hands change. The NMF algorithm was used to extract muscle synergies from the EMG data for push-up exercises. According to the results, two synergies were extracted from the EMG of 14 muscles. Synergy 1 activated at the lowest position. While synergy 2 contains the before and after the lowest position. Therefore, the most active muscles correspond to shoulder and elbow extension/flexion. Each synergy was controlled groups of muscles with similar biomechanical action and associated with the push-up phases. We have demonstrated the presence of muscle coordination in the push-up and examined their relationship with biomechanical variables. These results indicate how athletes and coaches can improve their performances by understanding the muscles synergies.

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